Mapping Magnetic Fields using Dust-Interacted Polarized Light

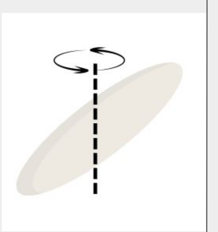
By: Rounak Bandyopadhyay

Background Research

Grain Alignment

Internal Alignment:

Cosmic rays act as stimuli, causing grains to spin unevenly, creating torque.



The grain reaches is lowest energy state (rotation around short-axis) via internal friction (Barnett Relaxation).

Grain has electric **dipoles** (mini-charge), and when rotating, these electric charges are **accelerating.**

Electricity (accelerating electric field) ⟶ **Magnetism**

Thus, grain is eventually aligned with galactic magnetic field due flipping electron spins (Barnett effect).

Short axis of grain is PARALLEL to magnetic field lines.

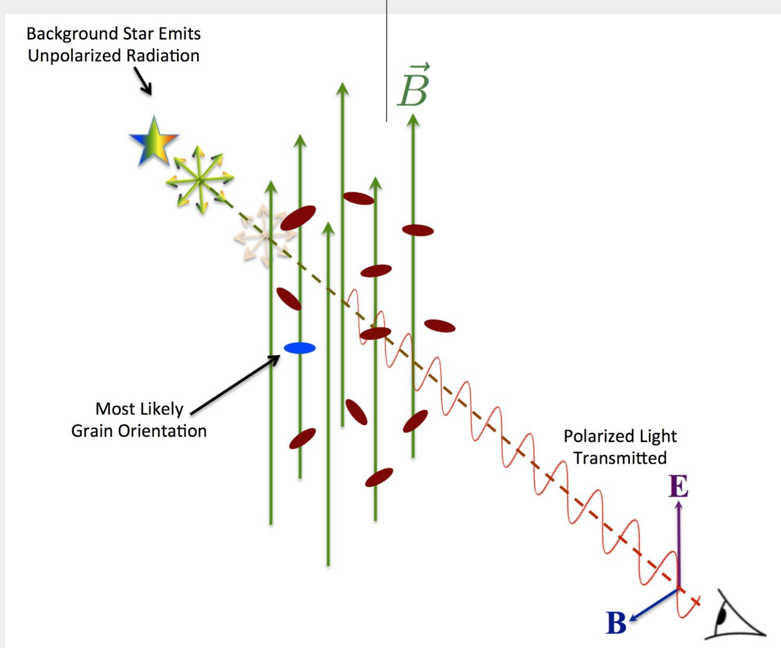


Figure 1. Steller Polarization due to grain alignment [2]

Stellar light **ABSORBED MORE** along long axis

Thus, light is **POLARIZED**, and electric field vector is **PARALLEL** to the short axis.

Problem/Question

Problem: While large scale maps of the magnetic field have been created and studied, smaller-scale maps looking for local anomalies or turbulence are often ignored.

Research Question: How do polarized stellar light vary along the Milky Way, and are there deviations or anomalies in small-scale magnetic maps in comparison to larger ones?

Hypothesis: Polarization vectors in small-scale regions will report local anomalies, turbulence, and unreported matter, which deviates in specificality from large-scale magnetic fields, which can lead to new discoveries of unreported matter.

Materials/Methods

Open Source Data Set: GPIPS DR4
- Near Infrared (H-Band) wavelengths
- Covered most of the Galactic Plane (galactic latitudes: −1° to +1° galactic latitudes: 18°–56°)

TOPCAT
- Java-based interactive data modifier and graphical analysis software tool

Methodology/Procedure

Step 1: Data Cleaning

Remove "Dirty" data:

Base Expression: UF==1 && !NULL_P && P!=0
- Only use high quality starlight (UF = 1)
- Remove all stars will P values of null or 0

4 Row Subsets made:

Clean 2 (Expression: UF==1 && !NULL_P && P!=0 && (GAL_L >= 36.5 && GAL_L <= 37.5) && (GAL_B >= -0.5 && GAL_B <= 0.5))
- Basic cleaning on 1° by 1° area

Clean 3 (Expression: UF==1 && !NULL_P && P!=0)
- Basic cleaning on whole dataset

Clean 4 (Expression: UF==1 && !NULL_P && P!=0 && (GAL_L >= 35.5 && GAL_L <= 37.5))
- Basic cleaning on 2° by 2° area

GPlane (Expression: UF==1 && !NULL_P && P!=0 && (GAL_L >= 36.5 && GAL_L <= 37.5) && (GAL_B >= -0.01 && GAL_B <= 0.01))

Step 2: Angle Calculation

Finding Polarization angle of starlight via Stokes vectors:

Formula for AngleRAD:

Θ = 0.5 * atan2(U, Q)

Stokes Parameters U and Q

Step 3: SkyVector Plotting

Plotting SkyVectors with aitoff projection:

Formula for DeltaLongitudeFINAL:
radiansToDegrees(cos(AngleRAD) / cos(GAL_B))

Formula for DeltaLatitudeFINAL:
radiansToDegrees(sin(AngleRAD))

Creates equal length vectors with correct angle direction.

Major Columns / Variables:

GAL_L: Galactic Longitude
GAL_B: Galactic Latitude
P: Debiased Polarization Percentage
SP: Uncertainty in Polarization percentage
Q: Stokes Parameter Q%
SQ: Uncertainty in Q value
U: Stokes Parameter U%
SQ: Uncertainty in U value
UF: Usage Flag (1 = High Quality)
DeltaLatitudeFINAL(Student-made synthetic column): Change in latitude for vectors
DeltaLongitudeFINAL (Student-made synthetic column): Change in longitude for vectors
AngleRAD (Student-made synthetic column): Angle of polarization vector in radians

Primary Column Table:

Table Browser for 1: GPIPS_DR4_Unique_Star_File.tbl

	NUM	GAL_L	GAL_B	P	SP	Q	SQ	U	SU	UF	DeltaLatitud...	DeltaLongitu...	AngleRAD
7769585	7769585	36.53077	0.29028	2.581	0.491	0.646	0.458	2.546	0.493	1	0.02978	2.12609	0.66115
7769612	7769612	36.8859	0.47241	2.2	0.596	0.504	0.574	2.223	0.597	1	0.02588	1.93047	0.67392
7769683	7769683	36.69023	0.3719	1.354	0.606	−0.68	0.601	1.319	0.608	1	0.02418	0.75642	1.0234
7769827	7769827	36.566	0.3079	1.177	1.277	−0.719	1.363	1.581	1.258	1	0.02052	0.66856	0.99881
7769829	7769829	36.73205	0.39308	1.831	0.423	0.178	0.434	1.871	0.423	1	0.02358	1.46648	0.73797
7769855	7769855	36.8512	0.45416	3.81069	0.9166	1.42937	0.89053	3.64944	0.92054	1	0.03982	3.50288	0.59875
7770109	7770109	36.57866	0.31391	2.23827	0.59276	1.04961	0.57377	2.06387	0.59758	1	0.02149	2.00602	0.55016
7770176	7770176	36.91802	0.48788	2.567	0.653	0.295	0.806	2.632	0.651	1	0.03269	2.16631	0.72959
7770310	7770310	36.85674	0.45622	2.12152	0.15201	−0.32584	0.1515	2.10186	0.15202	1	0.03193	1.53773	0.8623
7770510	7770510	36.82263	0.4384	2.803	0.768	−0.926	0.729	2.754	0.773	1	0.04635	1.80684	0.94758
7770530	7770530	36.57414	0.31089	4.02	0.164	1.934	0.167	3.528	0.163	1	0.03751	3.63312	0.53467
7770608	7770608	36.61725	0.33288	2.217	0.353	0.804	0.339	2.096	0.354	1	0.0233	1.93305	0.60226
7770643	7770643	36.90019	0.47797	2.149	1.318	−0.815	1.39	2.386	1.31	1	0.03563	1.40786	0.94997
7770668	7770668	36.70602	0.37833	1.231	0.824	−0.118	0.798	1.477	0.825	1	0.01773	0.89862	0.82526
7770734	7770734	36.84434	0.44919	0.52556	1.08339	0.21136	1.09318	1.18545	1.08308	1	0.00639	0.44729	0.69718
7770798	7770798	36.67766	0.36357	1.571	0.262	0.042	0.28	1.592	0.262	1	0.02117	1.20413	0.77221
7770858	7770858	36.6255	0.3367	1.978	1.152	1.066	1.163	2.025	1.149	1	0.01875	1.79411	0.54312
7770924	7770924	36.8947	0.47468	2.327	0.251	0.818	0.191	2.193	0.258	1	0.02465	2.14906	0.60689

Total: 13,861,329 Visible: 7,442 Selected: 0

Figure 2. Table created by student using TOPCAT

Mapped Vectors

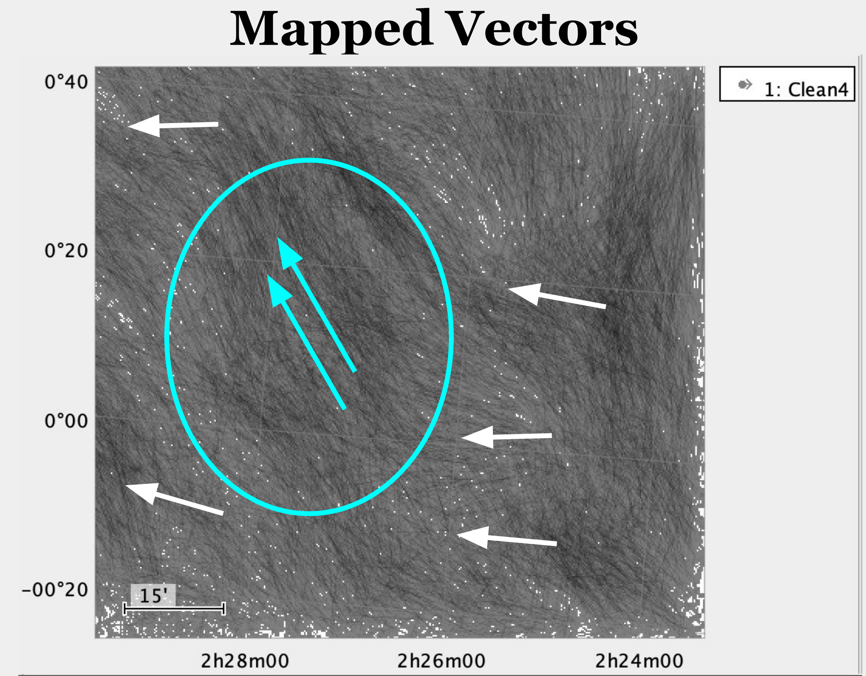


Figure 3. Small-scale SkyVector Plot created by student using TOPCAT

Local disturbances/turbulence detected. Vectors are at a whole in parallel direction to galactic field (horizontal)

Figure 5. Zoomed in Magnetic Map of Milky Way Polarized Emission created by Planck Satellite. *(Image credit: ESA and the Planck Collaboration.) [3]*

Near same area of sky as Figure 3. Slightly deviations upwards match, yet Planck map over-averages the turbulences.

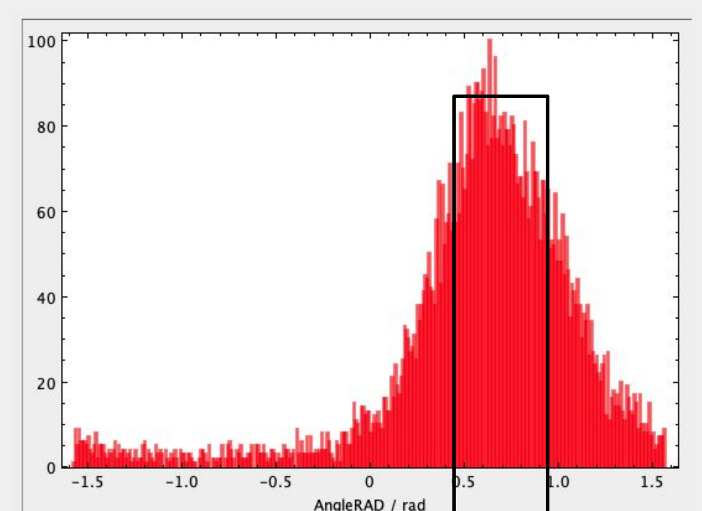


Figure 4. Histogram Graph created by student using TOPCAT

Most vectors show a light upward deviation

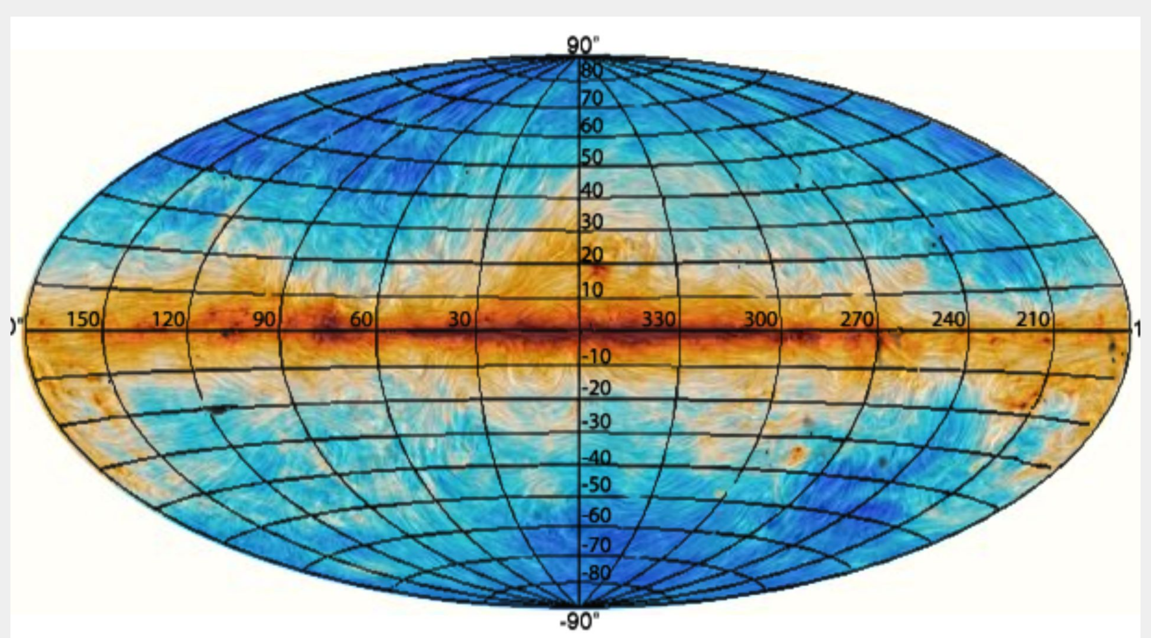


Figure 6. Magnetic Map using Milky Way Polarized Emission, created by Planck Satellite. *(Image credit: ESA and the Planck Collaboration.) [1]*

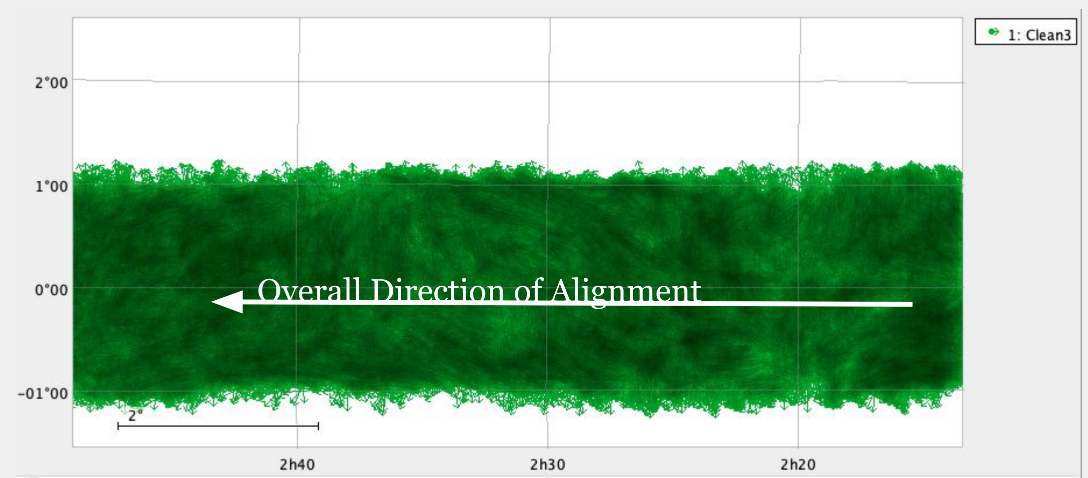


Figure 7. Larger-scale SkyVector Plot created by student using TOPCAT

Discussion

- As a whole, vectors are mostly **on average parallel** to galactic plane (matches predictions of Planck magnetic maps, and field-specific common knowledge)
  - Matches Planck magnetic maps in Figures 5, and 6.
- In Figure 3, local anomalies, **pointing upwards** is shown, matching slight upwards deviations in Figure 5. Also local anomalies of Figure 3 have **similar shapes and deviations** as Figure 7.
- In figure 4, vector angles tend to have a **upward deviation from galactic plane**, shown by majority of turbulent deviations in Figures 3,7, and 5.

Key Findings:

- Local Anomalies due to turbulence found, which are not fully mapped or shown (due to averaging vectors) at large scale maps
- While magnetic field at galactic plane is on average for the whole, directed along the plane, there are many turbulent factors that deviate from a straight line (Figure 7)

Conclusions:

My hypothesis was semi-accurate. This is because local anomalies due to turbulence was found that weren't clearly stated in larger scale maps. However my findings did not discover new matter of any sort specifically, while larger undetected matter could alter the galactic magnetic field, my research does not explicitly prove this.

Applications

Galactic magnetic fields is an ongoing field of research with many uncertainities, my mapping of this in a smaller scale could help build the foundation for find anomalies due to small, unaccounted turbulence.

With my pre-created expressions and synthetic columns, future research exploring different aspects of galactic magnetic field mapping could be used.

Essentially, my map creation helps test and refine galactic magnetic maps of the Milky Way.

Future Research:

There are multiple types of polarization/input data available:

- Thermal Emissions
- Zeeman Splitting
- Faraday Rotation
- Synchrotron Emission

I would like to create more maps for:

- Different data sources
- Different input styles (Zeeman Splitting, Thermal Emission, etc)
- Different part of the universe

Exploring different sources and maps could help find new perspectives (large/small scale) on the galactic magnetic field of the milky way. This is essential for research in this field, as smaller scales and latitudes away from the galactic plane are active areas of work and heavily underreachered.

Key References

Aghanim, N., Akrami, Y., Arroja, F., Ashdown, M., Aumont, J., Baccigalupi, C., Ballardini, M., Banday, A. J., Barreiro, R. B., Bartolo, N., Basak, S., Battye, R., Benabed, K., Bernard, J.-P. ., Bersanelli, M., Bielewicz, P., Bock, J. J., & Bond, J. R. (2019). Planck 2018 results. I. Overview and the cosmological legacy of Planck. *Astronomy & Astrophysics*. https://doi.org/10.1051/0004-6361/201833880

Andersson, B-G. (2016). *Dust Grain Alignment in the Interstellar Medium. | B-G Andersson – Astronomer*. Bgandersson.net. https://bgandersson.net/grain-alignment

Clemens, D. P., Cashman, L. R., Cerny, C., El-Batal, A. M., Jameson, K. E., Marchwinski, R., Montgomery, J., Pavel, M., Pinnick, A., & Taylor, B. W. (2020). The Galactic Plane Infrared Polarization Survey (GPIPS): Data Release 4. *The Astrophysical Journal Supplement Series*, *249*(2), 23–23. https://doi.org/10.3847/1538-4365/ab9f30

Planck Satellite Creates Detailed Map of the Milky Way's Magnetic Field. (2015, January 28). Dongascience.com. https://www.dongascience.com/en/news/6010